

# HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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ABSTRACT. A Hamiltonian fibration  $M \hookrightarrow P \rightarrow X$  (with monotone compact fibers) induces a kind of “bundle of  $A_\infty$  categories” over  $X$ , with fiber given by the Fukaya category of the fiber  $M$ . Algebraic  $K$ -theory ideas, together with the above picture for  $X = S^m$ , yield the following: if  $R$  is a regular ring and  $G$  is a compact Lie group then there is a natural group homomorphism  $\pi_m(BG) \rightarrow K_m(R) \oplus K_{m-1}(R)$ . Given a Hamiltonian fibration as above, the construction generally first assigns elements in a certain categorified algebraic  $K$ -theory of  $R$ , analogous to Toën’s secondary  $K$ -theory. Taking the Langlands dual of  $G$ , one may ask for a relationship between the images of the homomorphisms above. This is thematically the  $K$ -theory expression of electric-magnetic duality.

## 1. INTRODUCTION

One idea is that a Hamiltonian fibration over a smooth manifold  $X$  gives a kind of “bundle of  $A_\infty$  categories over  $X$ ”, with fiber the Fukaya category of the fiber. Exactly what this means is an interesting mathematical problem in its own right, and one answer is developed in [13]. Note that we are certainly not talking about local systems, as the “bundles” that one gets are generally non-trivial over spheres as shown in [11].

The second idea, due to Toën [18], is that there is a refinement of  $K$ -theory of a commutative ring  $R$ , by replacing  $R$ -modules with  $dg$  or  $A_\infty$  categories over  $R$ , and using Waldhausen  $K$ -theory. We will show that these two ideas naturally fit together. This culminates in a connection between symplectic geometry and algebraic  $K$ -theory, and in particular we get a certain geometric insight into what we call categorified algebraic  $K$ -theory of  $\mathbb{Z}$ . We are then naturally led to speculate on connections with Langlands duality phenomena.

Although in the main examples here the principal actors are compact groups and  $PU(n)$  in particular, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds; so we quickly introduce this. Let  $(M, \omega)$  be a symplectic manifold, throughout assumed to be compact, with  $\omega$  a closed non-degenerate 2-form on  $M$ . The group of diffeomorphisms  $\phi$  of  $M$ , satisfying  $\phi^*\omega = \omega$ , is called the symplectomorphism group of  $(M, \omega)$ . When  $M$  is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by  $\mathcal{H} = \text{Ham}(M, \omega)$ . In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields  $\{X_t\}$ ,  $t \in [0, 1]$ , satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions  $\{H_t\}$  on  $M$ .

The group  $\mathcal{H}$  is one of the more enigmatic objects of mathematics. It is always (unless  $M = pt$ ) an infinite-dimensional Fréchet Lie group with its  $C^\infty$  topology. But it has a natural, bi-invariant Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [8].

Much of the algebraic/topological structure of  $\mathcal{H}$  is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration*  $M \hookrightarrow P \rightarrow X$  is

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Supported by CONAHCYT research grant CF-2023-I.

a fiber bundle over  $X$ , with fiber a symplectic manifold  $(M, \omega)$ , whose structure group is contained in  $\text{Ham}(M, \omega)$ . For example, take  $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$  where  $\omega_{FS}$  is the Fubini-Study symplectic form, and take the structure group  $G = PU(n+1) \subset \text{Ham}(\mathbb{CP}^n, \omega_{FS})$ .

The other ingredient is algebraic  $K$ -theory, which is a construction associating an infinite loop space <sup>1</sup>  $K(R)$  to any ring  $R$ , and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called  $K$ -theory groups of  $R$ . This is a powerful tool in algebraic topology and ring theory, with applications in number theory and geometric topology.

Taking  $R$  to be the non-commutative ring  $\mathcal{K}$  of compact operators on a separable complex Hilbert space, the algebraic  $K$ -theory space  $K(\mathcal{K})$  <sup>2</sup> is identified with the topological  $K$ -theory space  $KU = BU \times \mathbb{Z}$ , see Karoubi [4], Suslin-Wodzicki [16]. In particular, a complex vector bundle  $E$  over a space  $X$  determines a homotopy class  $[f_E : X \rightarrow K(\mathcal{K})]$ , which we might call a *geometric element* of the corresponding abelian group. We shall see that in a certain sense this extends to a general ring  $R$ , provided it is commutative, and replacing a complex vector bundle by a suitable Hamiltonian fibration.

For each commutative ring  $R$ , we construct an infinite loop space  $K^{Cat}(R)$ . This is obtained by the Waldhausen  $K$ -theory construction on a certain category, whose objects are pretriangulated  $\mathbb{Z}_2$  graded,  $A_\infty$  categories over  $R$ , with a certain algebraic finiteness condition that we call finite type, see Definition 3.1. This categorified algebraic  $K$ -theory of  $R$  is a mostly minor variation of the secondary  $K$ -theory of  $R$  introduced by Toën.  $K^{Cat}(R)$  admits a natural map to the 2-periodic variant of algebraic  $K$ -theory of  $R$ , which we denote by  $K^{\mathbb{Z}_2}(R)$ . Technically,  $K^{\mathbb{Z}_2}(R)$  is the algebraic  $K$ -theory built using the exact category of 2-periodic complexes of  $R$ -modules.

*Remark 1.1.* It was pointed out to me by Bertrand Toën that for  $R$  a regular ring, for example  $R = \mathbb{Z}$ ,  $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$ . This can be proved with  $A^1$  homotopy theory.

To obtain “geometric elements”, we start with a Hamiltonian fibration  $M \hookrightarrow P \rightarrow X$ , with fiber a monotone symplectic manifold. If the Fukaya category of the fiber is finite type, then we show that for each  $R$ , the data of the isomorphism class of  $P$  determines a homotopy class  $[f_P : X \rightarrow K^{Cat}(R)]$ . As one consequence of this, we get:

**Theorem 1.2.** *There is a natural homotopy class  $[g : BPU(n) \rightarrow K^{Cat}]$  and so natural group homomorphisms:*

$$(1.3) \quad \pi_m(BPU(n)) \xrightarrow{g_*} K_m^{Cat}(R) \rightarrow K_m^{\mathbb{Z}_2}(R), \quad \forall m, n \in \mathbb{N},$$

for each commutative ring  $R$ . Furthermore, for any compact Lie group  $G$ , there is a homotopy natural map:

$$(1.4) \quad h : BG \rightarrow K^{\mathbb{Z}_2}(R).$$

*Example 1.* In the case of  $PU(2)$ , in [11], we outline the proof of injectivity of  $g_*$  for  $m = 4$  and  $R = \mathbb{Z}$ . We discuss a possible extension of this in Section 6. Injectivity for  $m = 2$  will be shown elsewhere. Also,  $K_1(\mathbb{Z}) = \mathbb{Z}_2$ ,  $K_2(\mathbb{Z}) = \mathbb{Z}_2$ , so assuming the Remark 1.1, we may expect that the induced map

$$h_* : \mathbb{Z}_2 \simeq \pi_2(BPU(2)) \rightarrow K_2(\mathbb{Z}) \oplus K_1(\mathbb{Z}),$$

is injective. Also  $K_3(\mathbb{Z}) = \mathbb{Z}_{48}$  and  $K_4(\mathbb{Z}) = 0$ . So given that  $g_*$  is injective for  $m = 4$ ,  $R = \mathbb{Z}$  we must conclude that the natural map  $K_4^{Cat}(\mathbb{Z}) \rightarrow K_4^{\mathbb{Z}_2}(\mathbb{Z})$  has a kernel.

<sup>1</sup>In this paper, the term infinite loop space is meant to be synonymous with connective  $\Omega$ -spectrum, i.e. a sequence of spaces  $\{E_n\}_{n \in \mathbb{N}}$  with  $E_n \simeq \Omega E_{n+1}$ . Then  $K(R)$  is the notation for the 0-space  $E_0$  of the corresponding spectrum.

<sup>2</sup>Rather one must take a certain non-connective variant of the algebraic  $K$  theory construction.

Taking the Langlands dual of  $G$ , we may ask about the correspondence of the respective mappings; we outline this in Section 8.

*Question 1.* Can one obtain the classes

$$[\mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R)]$$

by homotopy theory techniques, perhaps aided by some algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

## 2. PRELIMINARIES

Let  $R$  be a commutative ring. Denote by  $A_\infty \mathrm{Cat}_R$  the category of strictly unital,  $\mathbb{Z}$  graded  $A_\infty$  categories with morphisms  $A_\infty$  functors. We set  $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n}$  to be like  $A_\infty \mathrm{Cat}_R$  but with chain complexes  $\mathbb{Z}_n$  graded instead of  $\mathbb{Z}$  graded. This means that all structure maps, functors, etc., are with respect to the  $\mathbb{Z}_n$  grading. We will assume homological grading conventions on all chain complexes. Various subscripts  $R$  may be omitted.

There is an endofunctor

$$\mathrm{Perf} : A_\infty \mathrm{Cat} \rightarrow A_\infty \mathrm{Cat},$$

s.t.  $\mathrm{Perf}(B)$  is idempotent complete and pretriangulated for each  $B$ . One reference for this material is [2, Section 2.2]. We may denote by  $\widehat{C}$  the category  $\mathrm{Perf}(C)$ . This works similarly for  $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_2}$ , and this is the main subject of interest in this note.

Let  $\mathrm{Cat}_R$  denote the category of small  $R$ -linear categories. For  $C \in A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n}$ , we set  $hC \in \mathrm{Cat}_R$  to be the category with the same objects and with  $\mathrm{hom}_{hC}(a, b) = H_0(\mathrm{hom}_C(a, b))$ .

Given a morphism  $f : A \rightarrow B$  in  $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_2}$  we denote by  $f_* : hA \rightarrow hB$  the induced morphism in  $\mathrm{Cat}_R$ . We say that  $f$  is *quasi fully-faithful* if  $f_*$  is fully-faithful, and  $f$  is a *quasi equivalence* if  $f_*$  is an equivalence. We will say that  $f$  is a *Morita equivalence* if  $\mathrm{Perf}(f) : \mathrm{Perf}(A) \rightarrow \mathrm{Perf}(B)$  is a quasi equivalence.

## 3. BUILDING A WALDHAUSEN CATEGORY WITH OBJECTS $A_\infty$ CATEGORIES

To build an algebraic  $K$ -theory from the category of  $A_\infty$  categories, we need some analogue of “exact sequences” of  $A_\infty$  categories. For  $dg$  categories there are already very good answers to this, and this results in a definition of “secondary K-theory”, Toën [18]. The main difficulty in the case of  $A_\infty$  categories (with non-strict functors) is the existence of suitable colimits. In fact, strict colimits of diagrams in  $A_\infty \mathrm{Cat}$  do not exist. This is not surprising because the “correct” (in some geometric and algebraic sense, for instance Keller [5]) notion of functors of  $A_\infty$  categories, preserves  $A_\infty$  structures only in homotopy coherent sense, the latter is also discussed in Porter-Stasheff [9]. Despite this, there is a suitable theory of homotopy colimits, which will be sufficient for our purposes.

We also need to put some algebraic finiteness conditions on our  $A_\infty$  categories, otherwise any  $K$ -theory we construct will be trivial due to the Eilenberg swindle, the same reason that  $K$ -theory of infinite dimensional vector bundles is trivial. One natural idea is to require our  $A_\infty$  categories be smooth and proper or saturated in the sense of Kontsevich-Soibelman [7], as this is a dualizability condition, see [19]. We will work with a somewhat less restrictive notion, which naturally generalizes to the  $\mathbb{Z}_2$  graded case, and which is not too hard to check for geometric examples.

Set  $\mathcal{A} = \mathcal{A}_R \subset A_\infty \mathrm{Cat}_R^{\mathbb{Z}_2}$  to be the full-subcategory whose objects are pretriangulated and finite type, where this is defined as follows.

**Definition 3.1.** We say that  $B \in A_\infty \mathrm{Cat}_R^{\mathbb{Z}_2}$  is *finite type* if  $B$  is Morita equivalent to an  $A_\infty$  algebra, whose underlying chain complex  $B_\bullet$ , in each degree is projective and finitely generated over  $R$ . The same holds for the Hochschild homology complex of  $B$ .

It is shown in [3, Section A.4] that  $\mathcal{A}$ , has certain homotopy colimits, in particular for the pushout diagram:

$$(3.2) \quad \bullet \longleftarrow \bullet \longrightarrow \bullet.$$

*Remark 3.3.* Strictly speaking only distinguished homotopy cocones are constructed in [3], although they are expected to be homotopy colimits. Moreover, these distinguished cocones are suitably functorial, that is maps of diagrams induce natural maps of these distinguished cocones. The structure of distinguished functorial homotopy cocones is enough for the  $S$ -construction underlying the infinite loop space (4.1). This is a very minor generalization of the standard Waldhausen  $S$ -construction. To simplify the discussion, we will just call them homotopy colimits.

To define a Waldhausen structure on  $\mathcal{A}$ , we need to specify a class of morphisms  $\mathcal{C}$  called cofibrations and a class of morphisms  $\mathcal{W}$  called weak equivalences. We set these to be quasi fully-faithful maps and quasi isomorphisms respectively.

**Lemma 3.4.** *With respect to  $\mathcal{C}, \mathcal{W}$  as above,  $\mathcal{A}$  is a Waldhausen category. More specifically, we have:*

- (1)  $\mathcal{A}$  has a zero object.
- (2) The canonical morphism  $0 \rightarrow A$  is a cofibration for all  $A \in \text{obj } \mathcal{A}$ .
- (3) All isomorphisms are weak equivalences, and all weak equivalences are cofibrations.
- (4) Weak equivalences are closed under composition.
- (5) All diagrams of the form (3.2), where the right pointing arrow is a cofibration, have homotopy colimits.
- (6) Given a homotopy push-out diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow{g} & D, \end{array}$$

if  $f$  is a cofibration then so is  $g$ .

- (7) Given a diagram:

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{e} & C \\ \downarrow & & \downarrow & & \downarrow \\ D & \xrightarrow{g} & E & \xrightarrow{e'} & F, \end{array}$$

where  $e, e'$  are cofibrations, and vertical arrows are weak equivalences, the induced map of homotopy push-outs is a weak equivalence.

*Proof.* The zero object is defined to be the zero category (the category with unique object and morphism). Property 2 then holds vacuously. Property 3 and 4 hold by definitions of  $\mathcal{C}, \mathcal{W}$ . To check property 6 we can work directly with the construction in [3, Appendix A.4].

Specifically, if

$$F : J \rightarrow A_\infty \text{Cat}^{\mathbb{Z}_2}$$

is the functor representing the diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \\ C & & \end{array},$$

then first we form a type of Grothendieck construction  $\text{Groth } F$ , which is an  $A_\infty$  category, which we briefly describe. There are fully-faithful embeddings  $A, B, C \rightarrow \text{Groth } F$ , with images denoted by just  $A, B, C$ , s.t.

$$\text{obj } \text{Groth } F = \text{obj } A \sqcup \text{obj } B \sqcup \text{obj } C.$$

There is an identification

$$\text{hom}_{\text{Groth } F}(a, b) = \text{hom}_B(f(a), b),$$

for  $a \in A, b \in B$ . And a similar identification for  $\text{hom}_{\text{Groth } F}(a, c)$ . We call morphisms corresponding to the identity morphisms in this identification, *adjacent identities*.

These are all the non-zero  $\text{hom}$  sets, i.e. there is a non-zero morphism from  $o_1 \in F(i_1)$  to  $o_2 \in F(i_2)$  only if there is a morphism from  $i_1$  to  $i_2$  in  $J$ .

The homotopy colimit  $D$  is then defined to be the localization (an  $A_\infty$  analogue of the usual localization, as defined in [3]) of  $\text{Groth } F$  at the set of adjacent identity morphisms. There is then a homotopy commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow{g} & D. \end{array}$$

By construction, the induced commutative diagram:

$$\begin{array}{ccc} hA & \xrightarrow{f_*} & hB \\ \downarrow & & \downarrow \\ hC & \xrightarrow{g_*} & hD, \end{array}$$

is a push out diagram in  $\text{Cat}_k$ . As  $f_*$  is fully-faithful,  $g_*$  is fully-faithful by Stanculescu [15, Proposition 3.1].

The final property is verified in [3, Section A.4]. □

#### 4. CATEGORIFIED ALGEBRAIC K-THEORY

We define  $K^{\text{Cat}}(R)$  to be the infinite loop space given by the Waldhausen  $S$ -construction [20], on the Waldhausen category  $(\mathcal{A}_R, \mathcal{W}, \mathcal{C})$ .

More specifically,

$$(4.1) \quad K^{\text{Cat}}(R) = \Omega |wS_\bullet(\mathcal{A}_R)|,$$

where (very briefly)

- $S_\bullet(\mathcal{A}_R)$  is a simplicial category. For each  $d$ , objects of  $S_\bullet(\mathcal{A}_R)(d)$  are certain length  $d$  sequences of cofibrations in  $\mathcal{A}_R$  (with an additional structure.)
- $wS_\bullet(\mathcal{A}_R)$  is the simplicial category with  $wS_\bullet(\mathcal{A}_R)(d) = w(S_\bullet(\mathcal{A}_R)(d))$ , where the right-hand side is the category of weak equivalences of the category  $S_\bullet(\mathcal{A}_R)(d)$ .
- $|\cdot|$  is the geometric realization of a simplicial category, with the latter understood as a bisimplicial set, taking nerve of each  $wS_\bullet(\mathcal{A}_R)(d)$ .

**4.1. Some computation, and relation with Toën's secondary  $K$ -theory.** Toën [19], constructs what he calls the secondary  $K$ -theory spectrum  $K^{(2)}(R)$  of  $R$ . Our construction above is a certain modification of this construction. More specifically, Toën works with smooth and proper  $\mathbb{Z}$ -graded, pretriangulated  $dg$  categories. This can only be expected to hold for Fukaya categories of compact Calabi-Yau manifolds, and is not suitable for our geometric ambitions.

Define:

$$(4.2) \quad K_m^{Cat}(R) = \pi_m(K^{Cat}(R)).$$

To partially demystify this, we note that  $K_0^{Cat}(R)$  has the following description. By a cofiber sequence of  $A_\infty$  functors we mean a diagram in  $\mathcal{A}_R$ :

$$(4.3) \quad A \xrightarrow{f} B \rightarrow C,$$

with  $f$  quasi fully-faithful, which completes to a homotopy pushout diagram in  $\mathcal{A}_R$ .

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C. \end{array}$$

Then  $K_0^{Cat}(R)$  is the quotient of the free abelian group generated by the set of isomorphism classes of objects in  $\mathcal{A}_R$ , by the subgroup generated by

$$\{[B] - [A] - [C] \mid \text{there is a cofiber sequence (4.3).}\}$$

Let us also recall the definition of  $K_0(R)$ . This is the quotient of the free abelian group, generated by the set of isomorphism classes of projective finitely generated  $R$ -modules, by the subgroup generated by

$$\{[B] - [A] - [C] \mid \text{exists an exact sequence } 0 \rightarrow B \rightarrow A \rightarrow C \rightarrow 0.\}$$

More generally if  $\mathcal{E}$  is an exact category, one defines  $K_0(\mathcal{E})$  to be the quotient of the free abelian group, generated by the set of isomorphism classes in  $\mathcal{E}$ , by the subgroup generated by

$$\{[B] - [A] - [C] \mid \text{exists an exact sequence } B \rightarrow A \rightarrow C.\}$$

It is well known that  $K_0(D^{perf}(Mod_R))$  is isomorphic to  $K_0(R)$ , (together with higher  $K$ -groups), where  $Mod_R$  denotes the category of projective finitely generated  $R$ -modules and  $D^{perf}(Mod_R)$  denotes the derived category of perfect complexes.

**Lemma 4.4** (Toën [19]). *Denote by  $Hoch_\bullet(C)$  the Hochschild homology complex of  $C \in A_\infty Cat_R$ . There is a natural map*

$$\begin{aligned} K_0^{(2)}(R) &\rightarrow K_0(D^{perf}(Mod_R)) \simeq K_0(R) \\ [C] &\mapsto [Hoch_\bullet(C)], \end{aligned}$$

and it is non-trivial, where  $[\cdot]$  denote the classes in the associated Grothendieck  $K$ -groups.

Toën's proof of this proceeds as follows. One shows that images of cofiber sequences of  $A_\infty$  functors, by the functor  $C \mapsto Hoch_\bullet(C)$ , give exact sequences in  $D^{perf}(Mod_R)$ , of the associated Hochschild complexes. This property of the Hochschild functor is discussed for example in Keller [5, Theorem 5.2]. Other interesting computations in degree 0 have been made by Tabuada [17].

There is an analogous map in the case of  $K_0^{Cat}(R)$  but in general we no longer have that  $K_0(R) \simeq K_0(D_n(Mod_R))$ , where  $D_n(Mod_R)$  denotes the  $n$ -periodic derived category, as there are certain corrections, see Saito [10]. We do have:

**Theorem 4.5** (Saito [10]). *For  $n$  even, the natural homomorphism:*

$$(4.6) \quad \psi : K_0(D_n(Mod_R)) \rightarrow K_0(R)$$

$$(4.7) \quad \psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H_i(V)],$$

is an isomorphism.

**Corollary 4.8.** *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

where  $\phi$  is the map induced by  $[C] \mapsto [Hoch_{\bullet}(C)]$ , is non-trivial.

*Proof.* Take  $C$  to be a  $\mathbb{Z}_2$  graded  $A_{\infty}$  algebra over  $\mathbb{Z}$  such that  $HH_0(C) = \mathbb{Z}$ ,  $HH_1(C) = 0$ ,  $HH_2(C) = \mathbb{Z}$ , where  $HH_{\bullet}(C)$  is the homology of  $Hoch_{\bullet}(C)$ . It is not hard to construct an example. As a geometric example take  $C = FH_{\bullet}(L_0, L_0)$  where the latter is the Floer chain algebra of the equator  $L_0$  in  $S^2$ , then the Hochschild homology  $HH_{\bullet}(C)$  is isomorphic to the singular homology  $H_{\bullet}(S^2, \mathbb{Z})$ , [14].

So we get

$$\psi([Hoch_{\bullet}(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

**Corollary 4.9.**  $[\widehat{Fuk(S^2, \omega)}]$  represents a non-trivial element in  $K^{Cat}(\mathbb{Z})$ .

One natural question is how to produce interesting elements in  $K_m^{Cat}(R)$  for higher  $m$ . We will show that there are somewhat exotic elements coming by way of symplectic geometry.

## 5. $|\mathcal{A}_R|$ AS A REPRESENTING SPACE OF A “FIELD THEORY”

We briefly review simplicial sets. We denote by  $\Delta$  the simplex category:

- The set of objects of  $\Delta$  is  $\mathbb{N}$ .
- $\text{hom}_{\Delta}(n, m)$  is the set of non-decreasing maps  $[n] \rightarrow [m]$ , where  $[n] = \{0, 1, \dots, n\}$ , with its natural order.

A simplicial set  $X$  is a functor

$$X : \Delta^{op} \rightarrow \text{Set}.$$

The set  $X(n)$  is called the set of  $n$ -simplices of  $X$ .  $\Delta^d$  will denote a particular simplicial set: the standard representable  $d$ -simplex, with

$$\Delta^d(n) = \text{hom}_{\Delta}(n, d).$$

**Definition 5.1.** For  $X$  a simplicial set,  $\Delta(X)$  will denote the *simplex category of  $X$* . This is the category s.t.:

- The set of objects  $\text{obj } \Delta(X)$  is the set of simplicial maps:

$$\Sigma : \Delta^d \rightarrow X, \quad d \geq 0.$$

- Morphisms  $f : \Sigma_1 \rightarrow \Sigma_2$  are commutative diagrams in  $s\text{-Set}$ :

$$(5.2) \quad \begin{array}{ccc} \Delta^d & \xrightarrow{\quad} & \Delta^n \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

The main construction of [13] has as input a Hamiltonian fibration  $M \hookrightarrow P \rightarrow X$ <sup>3</sup>, with  $(M, \omega)$  a compact monotone symplectic manifold, e.g.  $(\mathbb{CP}^n, \omega_{FS})$ . And the output is a functor

$$(5.3) \quad F_P : \Delta(X_\bullet) \rightarrow A_\infty \text{Cat}_R^{\mathbb{Z}_2},<sup>4</sup>$$

well defined up to concordance of functors as defined below. We may understand  $F_P$  as giving a kind of “field theory”.

**Definition 5.4.** Let  $Y$  be a simplicial set and  $C$  any category. Let  $F_i : \Delta(Y) \rightarrow C$ ,  $i = 0, 1$  be functors. A **concordance** of  $F_0$  to  $F_1$  is a functor  $\tilde{F} : \Delta(Y \times I) \rightarrow C$  s.t.  $\tilde{F}|_{\Delta(Y \times \{0\})} = F_0$  and  $\tilde{F}|_{\Delta(Y \times \{1\})} = F_1$ .

Then we obtain a functor:

$$(5.5) \quad (\hat{F}_P = \text{Perf} \circ F_P) : \Delta(X_\bullet) \rightarrow A_\infty \text{Cat}^{\mathbb{Z}_2},$$

again well defined up to concordance of functors, with image now consisting of pretriangulated idempotent complete  $A_\infty$  categories.

By construction  $\hat{F}_P$  takes all morphisms to quasi equivalences in  $A_\infty \text{Cat}^{\mathbb{Z}_2}$ . So we get an induced map:

$$(5.6) \quad \hat{\mathcal{F}}_P : |\Delta(X_\bullet)| \rightarrow |wA_\infty \text{Cat}^{\mathbb{Z}_2}|,$$

where  $|C|$  is shorthand for the geometric realization of the nerve of a category  $C$ .

The same construction applies to a concordance of functors, yielding a homotopy:

$$(5.7) \quad |\Delta(X_\bullet)| \times [0, 1] \rightarrow |wA_\infty \text{Cat}^{\mathbb{Z}_2}|.$$

Since  $X \simeq |\Delta(X_\bullet)|$ , we conclude that  $\hat{\mathcal{F}}_P$  induces a well defined class

$$[X \rightarrow |wA_\infty \text{Cat}^{\mathbb{Z}_2}|]$$

in  $\pi_0(\text{Maps}(X, |wA_\infty \text{Cat}^{\mathbb{Z}_2}|))$ , where  $\text{Maps}$  is the space of continuous maps. By slight abuse we denote this class by  $[\hat{\mathcal{F}}_P]$ .

To summarize, we obtain the following variation of the main theorem of [13].

**Theorem 5.8.** *Let  $M \hookrightarrow P \rightarrow X$  be a Hamiltonian fibration with fiber  $(M, \omega)$  a compact monotone symplectic manifold. Suppose that  $\text{Fuk}(M, \omega)$  is finite type. Then the assignment:*

$$(5.9) \quad [P] \mapsto [\hat{\mathcal{F}}_P] \in \pi_0(\text{Maps}(X, |wA_\infty \text{Cat}^{\mathbb{Z}_2}|)),$$

*is well defined, where  $[P]$  is the Hamiltonian isomorphism class of the fibration  $P$ .*

**Proposition 5.10.**  *$\text{Fuk}(\mathbb{CP}^n, \omega_{FS})$  is finite type, where the latter denotes the monotone Fukaya category, [14].*

*Proof.* Specifically, we take a concrete model of  $\text{Fuk}(\mathbb{CP}^n, \omega_{FS})$  with objects oriented Lagrangians Hamiltonian isotopic to the Clifford torus  $L_0$ . In this case it is automatic that  $\text{Fuk}(\mathbb{CP}^n, \omega_{FS})$  is quasi-isomorphic to the  $A_\infty$  algebra  $HF(L_0, L_0)$ , and finiteness follows.  $\square$

*Example 2.* More generally, given a compact Lie group  $G$ ,  $G/T$  is always monotone with respect to some Kirillov-Kostant-Souriau symplectic structure  $\omega_{KKS}$ . (The latter is well defined up to deformation equivalence, see for instance [6]). It is natural to expect that  $\text{Fuk}(G/T, \omega_{KKS})$  is finite type, the study of its monotone Lagrangians is carried out for example in [1].

<sup>3</sup>Since the construction in [13] is carried out on the universal level, we do not need any smoothness assumptions on  $P$  here.

<sup>4</sup>In [13] we work with  $R = \mathbb{Q}$  but this is only needed for inverting quasi equivalences. As we don't need to do this here, the construction works perfectly well over any commutative ring.

**Corollary 5.11.** *There is a natural map:*

$$(5.12) \quad cl : B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS}) \rightarrow |w\mathcal{A}_R|,$$

*well defined up to homotopy.*

*Proof.* Let  $\mathbb{CP}^n \hookrightarrow P \rightarrow B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS})$  be the tautological Hamiltonian fibration, that is the fiber bundle associated to the universal principal  $\operatorname{Ham}(\mathbb{CP}^n, \omega_{FS})$  bundle over  $B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS})$ . By the Example 2 we may apply Theorem 5.8 to get a well defined up to homotopy map:

$$(5.13) \quad B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS}) \rightarrow |w\mathcal{A}_R|.$$

□

**Definition 5.14.** For brevity we call the symplectic manifolds in Theorem 5.8 *finite monotone type*.

## 6. HAMILTONIAN ELEMENTS IN CATEGORIFIED ALGEBRAIC K-THEORY

Now recall the definition of the infinite loop space (4.1). There is a natural map, see [21, Remark 8.3.2],

$$(6.1) \quad |w\mathcal{A}_R| \rightarrow \Omega |wS_\bullet(\mathcal{A}_R)| = K^{Cat}(R).$$

Combining with Corollary 5.11 we get:

**Corollary 6.2.** *For each commutative ring  $R$ , there is a natural homotopy class of a map:*

$$B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS}) \rightarrow K^{Cat}(R),$$

*and so homomorphisms:*

$$\mathcal{U}_{m,*} : \pi_m(B \operatorname{Ham}(\mathbb{CP}^n, \omega_{FS})) \rightarrow K_m^{Cat}(R).$$

As a first step towards the computation of the image of  $h_m$  we know that the map

$$(6.3) \quad (\mathbb{Z} \simeq \pi_4(BPU(2)) \simeq \pi_4(B \operatorname{Ham}(S^2, \omega))) \rightarrow \pi_4(|w\mathcal{A}_{\mathbb{Z}}^{\mathbb{Z}_2}|, \widehat{\operatorname{Fuk}(S^2, \omega)}),$$

induced by the map  $cl$ , is injective, [11].<sup>5</sup>

Thus, if we knew that

$$(6.4) \quad \pi_4(|w\mathcal{A}_{\mathbb{Z}}^{\mathbb{Z}_2}|, \widehat{\operatorname{Fuk}(S^2, \omega)}) \xrightarrow{\mathcal{U}_{4,*}} K_4^{Cat}(\mathbb{Z})$$

was injective, we would obtain non-triviality of the image of  $\mathcal{U}_{4,*}$ . And we would have:

$$(6.5) \quad K_4^{Cat}(\mathbb{Z}) \text{ conditionally admits a } \mathbb{Z} \text{ injection.}$$

In general, there is no reason to expect  $\mathcal{U}_{m,*}$  to be injective, but  $m = 4$  is in the stable range of homotopy groups of  $BPU(2)$ . Here ‘stable’ means the range in which  $BPU(n) \rightarrow BPU$  injects on homotopy groups. It is thus tempting to conjecture that there is a similar stabilization phenomenon for the more abstract mapping:

**Conjecture 1.**

$$\pi_{2k}(|w\mathcal{A}_{\mathbb{Z}}^{\mathbb{Z}_2}|, \widehat{\operatorname{Fuk}(\mathbb{CP}^{n-1}, \omega_{FS})}) \xrightarrow{\mathcal{U}_{2k,*}} K_{2k}^{Cat}(\mathbb{Z}) \text{ is injective for } 2k \leq 2n.$$

<sup>5</sup> [11] does not work with pretriangulations, however pretriangulating does not change the relevant computation.

The above is partly motivated by [12, Theorem 1.2], which says that the topological stable range of  $BPU(n)$  is related to the algebraic stable range of certain rational coefficients quantum characteristic classes, related to the Fukaya category theory underlying the construction of the Hamiltonian algebraic  $K$ -theory elements.

It should be emphasized that the latter conjecture is of algebraic topological nature, there is no analysis involved in its formulation (assuming  $\text{Fuk}(\mathbb{CP}^n, \omega_{st})$  is already given). On the other hand the content and the proof of the injectivity of (6.3) does involve geometric analysis.

## 7. HAMILTONIAN ELEMENTS IN ALGEBRAIC $K$ -THEORY

It was one of our initial goals to obtain “geometric elements” of classical algebraic  $K$ -theory.

*Proof of Theorem 1.2.* We have a Waldhausen category  $Ch_R^{\mathbb{Z}_2}$  of 2-periodic complexes of projective finite rank  $R$ -modules. Its Waldhausen  $K$ -theory infinite loop space will be denoted by  $K^{\mathbb{Z}_2}(R)$ . We have the map:

$$\phi_* : K^{Cat}(R) \rightarrow K^{\mathbb{Z}_2}(R),$$

where  $\phi(C) = Hoch_{\bullet}(C)$  as in Section 4. In particular, we obtain homotopy natural maps:

$$\text{BHam}(\mathbb{CP}^n, \omega_{FS}) \rightarrow K^{Cat}(R) \xrightarrow{\phi_*} K^{\mathbb{Z}_2}(R), \quad \forall n \in \mathbb{N}.$$

Taking the composition with the natural map  $BPU(n+1) \rightarrow \text{BHam}(\mathbb{CP}^n, \omega)$ , this proves the first part of the theorem.

If we knew that the monotone Fukaya category  $\text{Fuk}(G/T, \omega_{KKS})$  is finite type, we would likewise get homotopy natural maps:

$$\text{BHam}(G/T) \rightarrow K^{Cat}(R) \xrightarrow{\phi_*} K^{\mathbb{Z}_2}(R).$$

This may not be known at the moment. However, we may directly construct the maps

$$\text{BHam}(G/T) \rightarrow K^{\mathbb{Z}_2}(R),$$

as follows.

Let  $P$  denote the associated  $G/T$  bundle over  $\text{BHam}(G/T)$ . The composition  $Hoch_{\bullet} \circ F_P$  gives a functor  $\Delta(X_{\bullet}) \rightarrow Ch_R^{\mathbb{Z}_2}$  taking all morphisms to quasi-equivalences. Again by the Waldhausen construction, there is an induced, natural up to homotopy map,  $\text{BHam}(G/T) \rightarrow K^{\mathbb{Z}_2}(R)$ . And this specializes to give Theorem 1.2 in the introduction, using the natural map

$$BG \rightarrow \text{BHam}(G/T).$$

□

## 8. LANGLANDS DUAL ALGEBRAIC $K$ -THEORY ELEMENTS.

It is very natural to try to understand how Langlands duality fits into the story of the mappings  $\pi_m(BG) \rightarrow K_m(R) \oplus K_{m-1}(R)$ . Denote by  $K^{\vee}$  the maximal compact subgroup of the Langlands dual of the complexification  $G_{\mathbb{C}}$ . Via Theorem 1.2, we have a pair of natural maps:

$$(8.1) \quad \phi_G : \pi_{\bullet}(BG) \rightarrow K_{\bullet}^{\mathbb{Z}_2}(R),$$

$$(8.2) \quad \phi_{K^{\vee}} : \pi_{\bullet}(BK^{\vee}) \rightarrow K_{\bullet}^{\mathbb{Z}_2}(R).$$

If  $G$  is simply laced it is not hard to see that the images of these maps are the same, as  $G/T = K^{\vee}/T$  and the images of  $G$  and  $K^{\vee}$  in  $\text{Ham}(G/T, \omega)$  coincide. More generally, we may conjecture that there is some correspondence between the images:

**Conjecture 2.** *There is a natural endomorphism  $h : K_{\bullet}^{\mathbb{Z}_2}(R) \rightarrow K_{\bullet}^{\mathbb{Z}_2}(R)$ , s.t.  $\phi_G = h \circ \phi_{K^{\vee}}$ .*

This is thematically the conjectural, rudimentary  $K$ -theory expression of the electric-magnetic duality.

## 9. ACKNOWLEDGMENTS

This work was initiated during my stay as a member at IAS in Spring 2024, and I am very grateful for all that was provided. Many thanks to Bertrand Toën, Jacob Lurie, Nick Sheridan and Chi Hong Chow for answering many background questions.

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